

League of Overwatch at Moloco (Hard)

시간 제한	메모리 제한	제출	정답	맞힌 사람	정답 비율
2 초	512 MB	393	209	163	55.068%

문제

Once in a while, Moloco employees partition themselves into two groups, and engage in a best-of-five series of the League of Overwatch. Since some pairs of employees are hardcore gamers and they have played together as a duo in the past, the company decided to split them for this friendly company-wide event to make the event less competitive but more enjoyable for newbies.

Formally, there are n employees at Moloco who are numbered as 1 through n .

There are m known pairs of employees (f_i, s_i) such that the employees f_i and s_i have played a duo game in the past -- they must belong to different groups for this event.

Given this information, determine whether it is possible to partition n employees to two non-empty groups such that each employee belongs to exactly one group and there is no pair (f_i, s_i) such that both f_i and s_i belong to the same group.

입력

First line contains two integers, n and m where $1 \leq n \leq 1,000,000$ and $1 \leq m \leq 1,000,000$.

The following m lines contain two integers where $i+1$ th line describes f_i and s_i where $1 \leq f_i, s_i \leq n$. It is guaranteed that $f_i \neq s_i$ for all i .

출력

Your output should be a single line that contains a string "POSSIBLE" or "IMPOSSIBLE" (quotes for clarity only).

예제 입력 1

3 4

1 2

2 3

3 1

2 1

예제 출력 1

IMPOSSIBLE

예제 입력 2

3 3

1 2

1 2

2 3

예제 출력 2

POSSIBLE

힌트

Example 2: $\{2\}$ and $\{1,3\}$ is a valid partition.)

출처

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- 문제를 만든 사람: [haden](#)
- 데이터를 추가한 사람: [doju](#), [seok9311](#)

알고리즘 분류

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- [그래프 탐색](#)
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